

Zero-shot patient-conditioned weight modulation does not personalise video-based seizure detection

Phase 1 result report | 2026-09-14 | commit 71c4e57 | WU-SAHZU-EMU-Video, 8-patient LOPO cohort

Summary

An amortised hypernetwork generates LoRA weight deltas for a frozen VSViG backbone, conditioned on a behavioural signature computed from a patient's unlabelled interictal video alone. Across all eight leave-one-patient-out folds, the patient's own signature produces **no measurable advantage over another patient's signature**. Zero of eight folds meet a criterion registered before the experiment was run, and 6 of 8 point estimates favour the wrong signature.

The result is not a failure of the signature: it identifies its own patient at 55.5% against a chance rate of 12.5%. It is a failure of the training design. The hypernetwork is trained on exactly the patients the frozen backbone was already fitted on, where the loss it minimises is near zero, and it is asked to amortise a mapping from a 128-dimensional signature to weight deltas from five examples of that mapping.

Headline: own-signature minus wrong-signature = -0.00036 AUC (exact sign-flip permutation $p = 0.062$, $n = 8$). Restricted to the six folds whose metric can resolve the effect, -0.00048 ($p = 0.062$).

1. What was tested

Eight LOPO folds over patients pat01-pat04 and pat06-pat09; six further patients were excluded for insufficient interictal exposure. Each fold trains a VSViG backbone on five patients under a fixed 50-epoch budget with last-5-epoch weight averaging and no checkpoint selection, then freezes it including BatchNorm buffers, then trains a 741,537-parameter hypernetwork with the Max-Pool Dynamic Cyclic Sampler on patient-homogeneous 50/50 batches, BCEWithLogitsLoss, AdamW with a 300-step warmup to $1e-3$ and cosine decay. B is zero-initialised, so the adapted model is exactly the baseline before the first gradient step - verified at run time on every fold.

Discrimination is measured as **within-recording** AUC. Pooling clips across recordings inflates AUC by +0.07 to +0.10 here, because recordings sit at different baseline score levels and carry different class mixes; a model can score one recording above another without ordering anything correctly inside either. Patients are weighted equally and pairs weight recordings within a patient.

2. The pre-registered criterion

A gain over the unadapted baseline is *not* evidence of personalisation: perturbing a frozen network's weights at all can help. Personalisation was therefore defined, before the experiment, as the patient's own signature standing apart from the distribution obtained with every other cohort patient's signature:

$$z = (AUC_{\text{own}} - \text{mean } AUC_{\text{shuffled}}) / \text{sd } AUC_{\text{shuffled}} > 2.0, \text{ with } AUC_{\text{own}} \text{ above the mean.}$$

Each fold's held-out patient is scored eight times: once with their own signature and once with each of the seven others. 64 evaluations in total.

3. Result

fold	baseline	own z	shuffled mean	shuffled sd	z-score	AUC pairs
pat01	0.6037	0.6166	0.6165	0.0015	+0.05	7,544
pat02	0.8638	0.8623	0.8624	0.0004	-0.44	11,450
pat03	0.8537	0.8537	0.8537	0.0000	n/a	41

pat04	0.7143	0.8571	0.8571	0.0000	+0.93	14
pat06	0.5257	0.4905	0.4919	0.0052	-0.28	1,206
pat07	0.9667	0.9474	0.9475	0.0003	-0.59	1,710
pat08	0.7139	0.7186	0.7189	0.0009	-0.32	9,251
pat09	0.1799	0.1882	0.1891	0.0005	-1.61	1,801

Shaded rows are folds whose metric cannot resolve the effect under test (section 4). No fold reaches $z = 2.0$.

4. Two folds are not evidence

Within-recording AUC counts only recordings holding both classes. For pat04 that is a single recording with 7 ictal and 2 interictal clips - **14 ordered pairs**, so its AUC moves in steps of 0.071. For pat03 it is 41 pairs, steps of 0.024. The effect under test is roughly 0.0005. This is why every condition returned an identical number with a standard deviation of zero: the instrument cannot move. pat04's apparent +0.143 from adaptation is **two pairs changing order**.

This qualifies more than the present table. The cohort baseline reported in Phase 1 - a held-out mean of 0.678 across eight patients - includes both of these folds.

5. Statistical verdict

Sample sizes here are too small for a normal approximation, so these are exact paired sign-flip permutation tests over all 2^n sign assignments.

quantity	all 8 folds	6 resolvable folds
own & minus; shuffled (personalisation)	-0.00036 p = 0.062 2/8 positive	-0.00048 p = 0.062 1/6 positive
own & minus; baseline (any adaptation)	+0.01409 p = 0.719 4/8 positive	-0.00503 p = 0.594 3/6 positive

Two conclusions follow. Personalisation is **absent** - not merely non-significant, but negative in point estimate under both analyses, with 0 of 8 folds clearing a threshold fixed in advance. And the apparent benefit of adapting at all (+0.0141 across eight folds) is carried entirely by pat04's two-pair flip: among the six folds that can resolve anything it is -0.00503, slightly harmful.

6. The stated hypothesis

Section 3.5 of the methodology predicts that the benefit of personalisation correlates *positively* with how atypically a patient moves relative to the cohort, measured as $D_p = \|z_p - c\|$ with c the centroid over that fold's training patients.

correlation with D_p	all 8 folds	6 resolvable folds
own & minus; baseline (as §3.5 defines benefit)	+0.049	-0.495
own & minus; shuffled (personalisation only)	-0.566	-0.706

Every version on the resolvable folds is negative. At $n = 6$ to 8 none is significant; the honest statement is that the predicted relationship is not present and the data lean the other way.

7. Why it failed

The diagnosis matters more than the negative, because it is a property of the design rather than of this run.

7.1 The objective is already satisfied

The methodology trains the backbone on five patients, freezes it, then trains the hypernetwork on *those same five patients*. Measured on fold pat01, the frozen backbone's BCE is **0.0066** on its own training patients against **0.726** on unseen ones - a 112-fold gap. The loss the hypernetwork minimises is near zero on every example it is permitted to see, while generalisation to an unseen patient never enters the objective. Training behaves accordingly: training BCE falls toward $1e-5$ while validation BCE rises monotonically.

7.2 Five examples of the mapping

A hypernetwork learns a function from a 128-dimensional signature to weight deltas. With five training patients it has five examples of that function. This is a sample-size problem, not an optimisation problem, and it predicts what was observed: deltas that saturate their Frobenius budget and then vary with the signature in a way unrelated to which patient it belongs to.

7.3 Not the signature's fault

The signature carries real patient identity - 55.5% identification accuracy across the eight backbones (range 46.9% to 62.5%), against 12.5% chance, every fold significant by permutation. But identification recall correlates **+0.726** with baseline AUC: the signature is least recoverable for exactly the patients the model most needs help with. It also rests mostly on the mean of stage-2 features rather than their variability, which leaves open whether it encodes motor behaviour or static appearance.

8. Threats to validity

- **Cohort size.** Eight evaluation patients and 18 seizures. Every correlation reported here is underpowered; the sign tests are exact but weak.
- **Measurement resolution.** Two of eight patients cannot support a clip-level AUC. Event-level metrics would be the appropriate instrument for them and are not yet reported alongside.
- **One optimisation family.** Two regimes were tried (peak LR $1e-3$ with a 0.5 delta budget, and $1e-4$ with 0.1) and agreed. A wider search was not run.
- **The baseline is not clinically usable.** At per-fold thresholds selected on internal validation patients it detects 18 of 18 seizures at 41 false alarms per hour. Conclusions here concern discrimination, not deployment.
- **Batching asymmetry.** The adapted arm uses patient-homogeneous batches with frozen BatchNorm; the baseline was trained with shuffled batches and live BatchNorm. A baseline retrained under the adapted arm's batching would make the comparison exactly like-for-like.

9. What would be required to test the method fairly

The result above tests the method as published. It does not establish that amortised patient conditioning cannot work, because two design faults - a saturated objective and five amortisation examples - are confounded with the method itself. A fair test needs the hypernetwork to train on patients the frozen backbone has never seen, and needs more than five of them. Both are addressable with data already held.

Reproducibility

Every figure is computed from files under `outputs/` by `scripts/build_report.py`. The full decision record, including the errors found and corrected during this work, is in `DECISIONS.md` (D1-D30). Each result file carries a run manifest with its seed, git commit and configuration.